

# Database Normalization

Normalization organizes relational databases to reduce redundancy. 1NF eliminates repeating groups. 2NF removes partial dependencies on composite keys. 3NF removes transitive dependencies. BCNF is a stricter form of 3NF. SQL joins (INNER, LEFT, RIGHT, FULL) combine rows based on related columns. Indexes speed up query performance. Transactions follow ACID properties: Atomicity, Consistency, Isolation, Durability.